

How do rising interest rates influence portfolio allocation between equities, bonds, and alternative assets?

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Abstract

Interest rates play a central role in determining the financial market's behavior. They affect the cost of borrowing money, the decisions surrounding investment, and the value of assets. Interest rates, across the world, have moved into a rising phase because of the actions taken by central banks to control inflation. Rising interest rates have a profound effect on the way investors allocate their portfolios to various asset classes.

The primary research question in this study was to better understand how changes in interest rates will affect investor portfolio allocations. The study looks specifically at the ways in which investors reevaluate how they invest in equity, bond and alternative assets as interest rates change.

The research was conducted using qualitative research methodology, employing secondary data analysis, reviewing existing research literature, and performing an analytical discussion of both the current market trends and the theoretical frameworks related to investing.

It was found that, generally, rising interest rates create, higher yielding bonds, which make them more attractive for investors, and as a result, equities typically come under downward pressure due to higher discount rates and higher costs to borrow. Therefore, alternative assets have become increasingly important, as they provide greater diversification along with risk management.

This research has significant implications for both investors and policymakers, and suggests that both dynamic portfolio strategies should be employed by investors, and balanced monetary policies should be utilised by policymakers in order to maximize returns and minimize risk.

1. Introduction

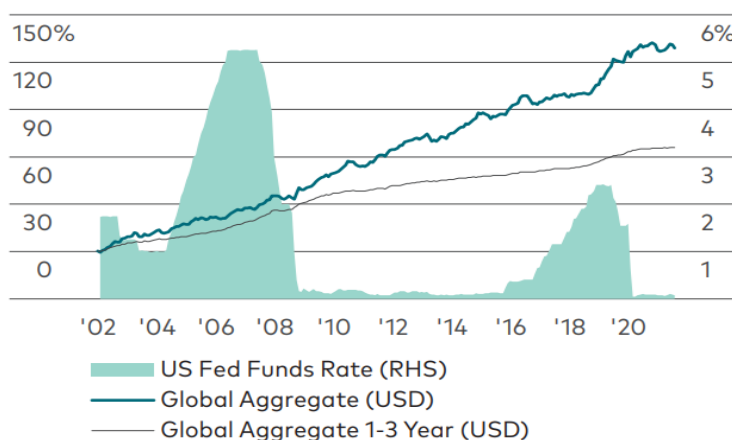
Interest rate changes are a critical component of both financial markets and macroeconomic conditions; they can impact lending costs for consumers and firms, as well as influence their decisions regarding investment and spending in the economy, thereby ultimately impacting overall economic activity. Central banks use increases and decreases in interest rates as a principal policy tool for controlling inflation and stabilising economic growth. Additionally, changes in interest rates can directly influence asset prices as they will change the discount rates used by investors when determining the future value of various assets, as well as the expected rate of return or the liquidity condition of financial market players that will be buying and selling these assets.

The connection between asset prices and central bank policy through interest rate changes has been very important since many companies have had to lower their anticipated earnings due to the impact of higher interest rates on borrowing costs and, therefore, reduced profit margins,

and many equity values are now lower than they would have been without the current higher level of interest rates. Also, rising interest rates typically lead to higher bond yields, making them more attractive when compared to equities for investors; thus, in most cases, when central banks are increasing the interest rate, global financial markets are being disrupted, as well as their investors.

From this perspective, portfolio allocation decisions become instrumental when determining where to invest funds to optimise the desired balance of risk and reward in today's global financial markets. In order to make better informed investment decisions within the current global financial markets, understanding how different asset classes behave in relation to interest rates will assist those making these decisions. This motivates our research into understanding how increasing interest rates have impacted and will continue to impact the behaviour of investors across the various asset classes.

Figure 1. A rising cash rate does not necessarily mean negative returns for bonds



U.S. Fed Funds rate change and Cumulative Performance of Global Aggregate and Global Aggregate 1-3 Year.

Source: Bloomberg. Data as at 31 March 2021.

The research also seeks to answer:

- How have rising interest rates impacted the performance of equities, bonds, and alternative assets?
- What changes do investors make to their portfolios as a result of rising interest rates?
- What variables impact the reduction of capital allocation in one asset class and increase capital allocation in another?

2. Conceptual Framework

Interest rates represent the cost you incur when borrowing money or the return you receive when lending money, and they can be determined by either central bank policy rates or market yields as determined by the forces of supply and demand in the marketplace.

Portfolio allocation is a process by which investments are strategically allocated among asset classes in order to create an optimal balance between risk and return. The primary asset classes are equities (equities), bonds (fixed income securities), and alternative assets (like gold, real estate, and hedge funds), each with its own risk and return characteristics, and different sensitivities to changes in the economy.

The two primary concepts that will guide the decision-making process surrounding portfolios are the risk-return trade-off and diversification; the risk-return trade-off says that there will be a higher expected return from an investment when the risk is higher, while diversification helps to reduce the total risk of a portfolio by distributing the investments among different types of assets that have varying levels of risk associated with them. The research for this paper is based on the Modern Portfolio Theory developed by Harry Markowitz, with its focus on creating

an efficient distribution of investments to optimize the return of a portfolio given its level of total risk, and a Capital Asset Pricing Model (CAPM) developed to describe the relationship between the systematic risk of a portfolio and its expected return. Finally, the concept of duration relates to the sensitivity of a bond's price to changes in interest rates, with an emphasis on how the duration of long-term bonds will magnify the effects of interest rate changes on the bond price.

Linkage:

- Rising rates → higher bond yields → equity valuation pressure → shift in allocation

3. Review of Literature

Numerous studies have established that stock market returns are heavily affected by the movement in interest rates. For example, research has shown that whenever there are increases in the interest rate, it negatively impacts the valuation of equities due to a higher discount rate and a higher cost of using capital (Ioannidis & Kontonikas, 2008). Recently obtained international evidence has shown that countries with larger increases in the yield on government bonds underperform (in terms of stock returns), therefore confirming that there is an inverse relationship between interest rates and stock market returns (Jordà et al., 2023). However, it is important to note that a consistent mapping of this relationship across countries and time frames does not exist, indicating that the macro and micro economic conditions have a strong influence on their relationship.

In the bond market, there is consistent research that suggests that there is an inverse correlation between bond price and interest rate, with higher prices leading to a larger yield and a smaller capital loss on existing bonds, therefore providing an incentive for investors to rebalance

their portfolios toward fixed income securities during tightening periods (Zheng, 2025). Also, the stock-bond relationship is essential to portfolio management due to the influence of a change in correlation on each of these elements of the portfolio, creating a new strategy for the investor (Portelli & Roncalli, 2024). Furthermore, recent studies have shown that due to the pandemic, many of the traditional correlations that existed among the various asset classes have changed significantly and therefore new allocations will need to be made dynamically in order to maximise investment returns (BlackRock, 2024).

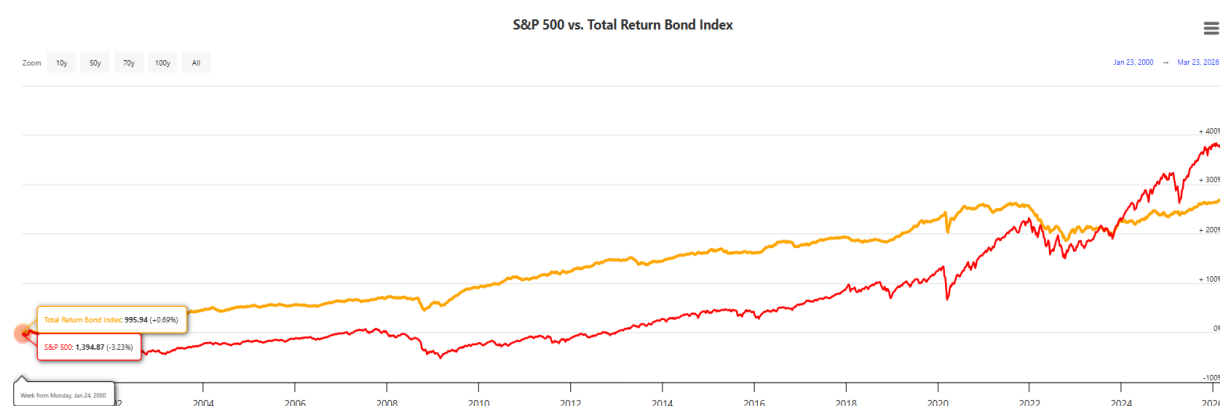
There has also been an increased interest in the use of alternative assets, primarily during periods of increasing rates and volatility. In periods of time when the market is volatile, real estate, commodities and private assets are often seen as an excellent source of diversification due to their relative lack of correlation to more traditional types of assets.

There is widespread empirical evidence of the negative correlation of equity values with increasing interest rates. Rising rate environments decrease the present value of cash flows from future earnings thereby reducing investor appetite for equity investment. However, rising interest rates produce a positive impact on bond yields making fixed income investments increasingly appealing to income-oriented investors. Nevertheless, the findings are not entirely consistent. Some of the studies have cited various market segments like banking and financial services sectors as possibly benefiting from an increase in interest rates because of the increase in their net interest margins illustrating that performance does vary within market segments.

Despite the ample literature that exists there still exists a critical research gap. Most empirical studies tend to analyse the performance of equities, bonds, or alternative asset classes in a vacuum. There are very few comprehensive studies that analyse and consider how rising

interest rates affect the allocation of capital among all three asset classes. Accordingly, this demonstrates the need for an integrated framework that captures the interrelatedness and dynamic shifts in investor behaviour as they relate to diverse capital allocations across a diversified portfolio.

S&P 500 vs. Total Return Bond Index



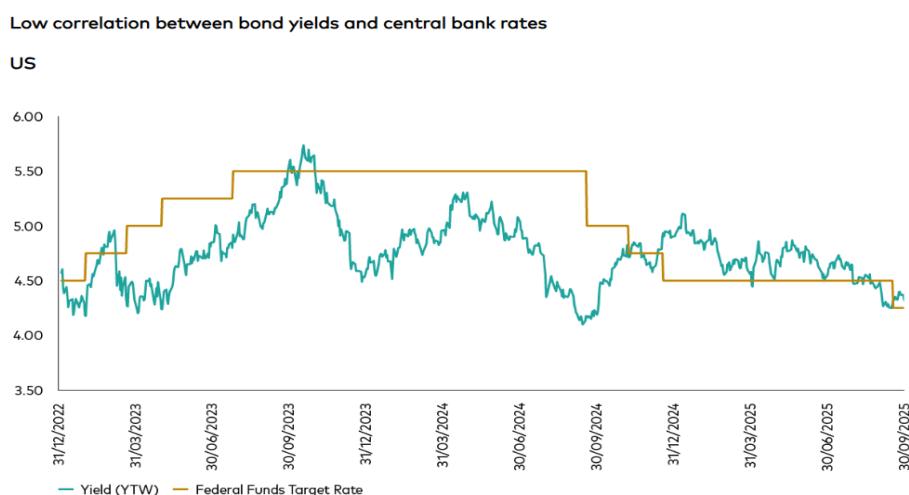
Source: [LongTermTrends](#)

4. Transmission Channels of Interest Rate Changes

Financial markets and portfolio allocations are impacted by interest rate changes in several ways through various channels that affect the prices of assets, the behaviour of investors, and the decisions of capital allocations. One primary channel through which interest rates affect the amount of capital available is the cost of capital. As interest rates rise, it costs more to borrow money, and as such, companies and individuals will reduce their ability to borrow. With fewer loans issued, corporate investment is lower and expectations for profit are lower; therefore, companies' market cap or value is also lower, and therefore much less likely to take on risky investments.

Another major channel is known as the discount rate impact. The present value of future cash flows determines how much of today's assets are valued. If interest rates rise, the discount that will be used to determine the value of those future cash flows will increase, thus decreasing the current valuation of the expected cash flows. Similarly, if interest rates rise, the yield on new fixed income investments will also increase, and thus those new fixed income investments will become more attractive to investors.

Another channel in which interest rate changes can impact the financial markets and the portfolios of investors is the availability of liquidity. When the economy has higher interest rates, it decreases the amount of money circulating within the economy, thereby restricting banks and credit unions from lending to others. Therefore, this has a negative impact on the volumes of trading in the markets and, ultimately, on the amount of money available to participate in the financial markets and buy assets. As a result of higher interest rates, investors will often then look to reduce their risk by moving from higher risk assets like stocks to lower risk assets like bonds.

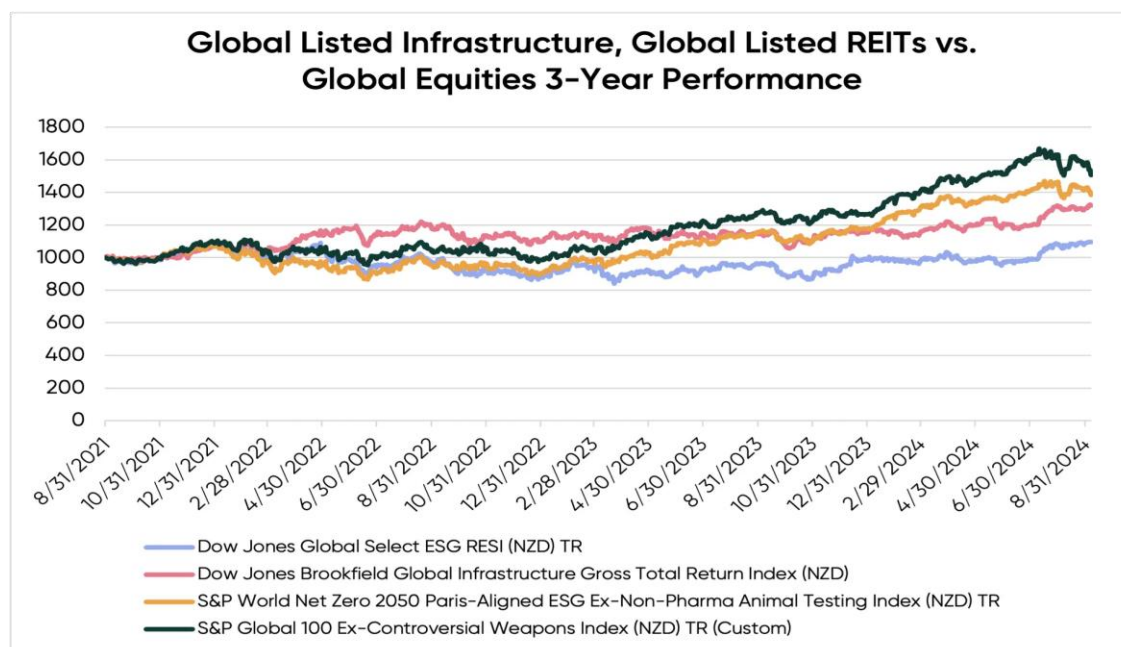


Source: [Vanguard](#)

5. Impact on Equities

Equity markets are impacted differently by rising interest rates based on differences in sectors as they relate to their business models, capital structures and exposure to the economy.

The sector-by-sector impact of rising interest rates is described below:



Source: [KernelWealth](#)

5.1. Technology & Growth Stocks

These sectors are vulnerable to changes in the discount rate because they rely heavily on estimated future earnings. When interest rates increase, the present value of cash flows expected to be generated from investments is reduced. As well, their cost of financing through third parties becomes much more expensive. Therefore, the result is a significant reduction in valuations and greater volatility associated with these sectors.

5.2. Banking & Financial Services

The banks & financial services sector benefits from the increased net interest margins of the lending activities. The bank earns more on the loan issued than what it pays on the deposit. Thus, profits generated from their lending operations will increase, thereby resulting in increased earnings and stock prices relative to other sectors.

5.3. Real Estate & Infrastructure

Real estate & infrastructure sectors are highly dependent upon debt to finance their operations; therefore, as interest rates increase, the cost of financing projects and obtaining mortgages will increase. In consequence, demand for the purchase of homes and infrastructure will decline significantly, resulting in lower sales and selling pressures on their stock prices.

5.4. Consumer Discretionary

With increasing interest rates, consumers ability to spend decreases. As the rate of increase in equated monthly installments (EMI's) increases, the demand for consumers to buy discretionary items (i.e., non-essential items) will decrease. As a result, revenues will decrease, and growth will slow, for businesses in the consumer discretionary sector.

5.5. Consumer Staples

Demand for products classified in consumer staples will remain constant, as these goods are considered "necessary". Therefore, this sector will be less affected by the economic

environment than any other sector, and consequently, the consumer staples sector will be less volatile.

5.6. Industrials & Manufacturing

The industrials and manufacturing sector is experiencing a moderate negative impact. As the cost of borrowing increases, the amount of money that can be invested into adding new capacity in this area. As a result, the level of economic activity is decreasing and therefore, will result in a lower level of production from this sector than would otherwise happen. Therefore, the performance of the sector is expected to decline moderately.

5.7. Energy & Commodities

Energy and commodities are impacted by global demand and the price of commodities. The current inflation-environment would support the price of energy and commodities so they are in a good position to perform and have a good chance of performing well even as interest rates increase.



Source: [Cutcher](#)

6. Impact on Bonds

Interest rate changes often directly affect the stock markets more so than the bond markets, but both have an indirect effect on one another.

Rising interest rates create more opportunities for investors to diversify their portfolios into more diversified, higher yielding assets. At the same time, rising interest rates create an opportunity for the investor to re-position their existing fixed income and cash assets into new asset class opportunities that yield a higher rate of return.

Interest rates have long been a significant driver of the global economy and continue to affect the stock and bond markets globally. Interest rates can have both a positive and negative impact on various asset classes and equities as they relate to each other, but knowing how they will fluctuate is one of the most difficult things to do in the investment process.

A key concept in regard to the duration effect is the sensitivity of a bond to changes in interest rates. Long-duration bonds (i.e., those with long maturities) have greater sensitivity to fluctuations in interest rates; therefore, they tend to suffer of a loss of capital when interest rates increase. In comparison, short duration bonds are not as sensitive to changes in interest rates because they mature faster and can be invested again quickly at a potentially higher yield.

Generally, while rising interest rates can cause capital losses in the short term for bond portfolios, they will improve long-term income potential through the reinvestment of the proceeds at higher rates of return.

Rolling Three-Year Stock/Bond Correlations



Source: [Morningstar](#)

7. Impact on Alternative Assets

7.1 Gold

Gold is negatively correlated with interest rates and is negatively impacted by rising interest rates because it becomes more expensive to hold an asset like gold that does not produce an income (i.e., yields), as well as when investors are looking for products that provide income (e.g., bonds). The same is true when there are periods of inflation or uncertainty.

7.2 Real Estate

Real estate is expected to be negatively impacted by rising interest rates. Because the industry is so dependent on loans to finance the purchase of real estate, when interest rates go up, it increases the cost of obtaining a loan, thereby making it less affordable to

buy property. Increased costs to obtain financing will have a dramatic impact on real estate developers and investors, resulting in a slowdown in the growth of prices in the real estate sector.

7.3 Hedge Funds and Commodities

Hedge funds and commodities are typically used to be a source of diversification in a portfolio because they have low volatility and are not related to traditional asset classes. Hedge funds and commodities have a chance to outperform even if there is an overall increase in interest rates. Hedge funds use various advanced techniques for risk management including hedging and arbitrage. They also create a moat around the volatility associated with equities and bonds in an investor's portfolio.

7.4 Cryptocurrency

Cryptocurrencies are very sensitive to liquidity conditions and monetary policy changes. An increase in interest rates decreases the excess liquidity available to invest in cryptocurrencies and thus, the price of cryptocurrencies. As a result, cryptocurrencies are viewed as a very high-risk, high-speculative asset class that has much greater price volatility when compared to traditional investments.

8. Portfolio Rebalancing Behaviour

The impact of rising interest rates on investor allocations create changes in portfolio allocations in order to maintain the greatest risk/return. One of the most common responses is to move from equities to bonds. When interest rates rise, newly issued bonds have greater yields than previously issued bonds, making bonds much more attractive than equities to investors who

want to invest in a low-risk, steady income strategy. This create a shift in capital flows from equity securities into fixed income securities.

Investor behavior often changes during periods of increasing interest rates, with many investors allocating more of their portfolio to defensive assets like utilities, gold, hedge funds, commodities and consumer staples. These types of investments tend to be less impacted by changes in the economy or interest rates and can help investors achieve consistent returns over time, even when monetary policy is contracting or the equity markets are unstable. As a result, the overall risk of an investor's portfolio is reduced with the portfolio's allocation to these types of investments.

Diversification into alternative investments (such as gold, real estate, hedge funds, and commodities) is another trend in investor behavior during periods of increasing interest rates. Alternative investments can provide the benefits of lower correlations to traditional asset classes and can help protect against inflation and market volatility, contributing to the overall stability of a portfolio.

When making portfolio decisions in a rising interest rate environment, investors are increasingly focused on risk-adjusted return analysis. Investors are more cautious about the trade-offs between risk and return, and often prefer to hold assets that provide higher expected returns for a corresponding level of risk. This potentially leads to more dynamic use of portfolio rebalancing strategies to preserve capital while maximizing long-term portfolio returns.

9. Empirical Discussion / Analysis

Recent empirical data from the time period immediately following the 2020 outbreak of COVID-19 shows the significant impact of rising interest rates on the allocation to portfolios as well as the financial markets. As commercial banks globally increase their base rates to combat rising inflationary pressures post-COVID, central banks (particularly in the United States and Europe) have implemented aggressive rate increases, resulting in significant changes in both the price of an asset as well as how an investor holds it.

9.1 Evidence from Recent Rate Hike Cycles

- By raising interest rates, the costs of capital and stock valuations fell for both developed and emerging economies.
- The bond market experienced initial volatility, but higher yields have made bonds more attractive as an alternative investment (OFI Invest. 2024).
- Changes in the interest rate had a major effect on investment strategies, borrowing capacity and perceptions of risk.

9.2 Comparison: Developed vs Emerging Markets

- **Developed markets:**
 - The effects of monetary policy on asset prices are transmitted to markets much quicker.
 - Equity markets are much more responsive to changes in interest rates.

- **Emerging markets:**
 - Many emerging economies demonstrated a greater level of resiliency compared to developed economies as a result of structural factors, as well as flexibility with regards to their respective monetary policies (IMF, 2024).
 - Emerging markets remain vulnerable to spillovers from tightening US monetary policy (Lastauskas & Nguyen, 2024).
- The data supports an influx of capital into emerging market debt securities, primarily when global yields have risen and investors seek returns greater than what is available in their domestic markets.

9.3 Role of Key Factors

1. Inflation Expectations

- Expectations of increasing inflation will result in tighter monetary policy and therefore contribute to increasing rates.
- Equity valuations will decline and demand shift from equities to bonds and real assets as the expected rate of inflation increases.

2. Monetary Policy Stance

- The monetary policy of a country's central bank can influence equity and bond markets, through both hawkish policy (tightening policy) leads to declines in equities and rising yields on debt; changes in the monetary policy of countries can also influence global capital flows and the value of countries' currencies relative to each other.

10. Policy and Practical Implications

National and international central banks are facing historically high commodity price volatility due to the COVID-19 pandemic and the rising costs of energy and food. This volatility will be compounded by increasing supply chain disruptions throughout the world caused by the Ukraine war and China's strict zero tolerance for COVID-19. As a result of these events, rising inflation will lead central banks to raise interest rates aggressively worldwide. In addition to the negative impacts of rising inflation on family and business budgets, the rapid interest rate increases coming from multiple central banks will continue to disrupt the prices of commodities and the economies of all countries. The financial markets impacted by rising commodity prices, inflation, and aggressive rate increases will require the U.S. government and all developed nations through other coordinated central bank efforts to restore the global economy's growth potential. All financial markets globally depend on coordinated monetary and fiscal policies to maintain their orderly operation. The financial markets must operate within a framework of agreed-upon central bank policy objectives to assist federal and state governments at all levels in supporting the wellness of their citizens and businesses, as well as restoring and maintaining the orderly operation of global financial markets.

11. Conclusion

The research demonstrates that changes to interest rates impact portfolio allocation among the main asset classes. The results show that as interest rates grow, equities tend to become less attractive because of the increase in their discount rates or financing costs by increasing their corporate valuation, as well as lowering investors' willingness to invest into equities, or investor confidence. Fixed income returns will increase because of an increase to the

yield of bonds due to higher interest rates. With that being the case, risk averse investors will likely find fixed income to be a more appealing asset class because they will be able to earn return at a fixed rate with stable yields. Furthermore, the usage of other asset types are becoming more prevalent as investors search for diversification of their portfolios through the use of alternative assets such as gold, commodities and hedge funds in order to reduce total portfolio risk during times of market volatility.

The interactions of the aforementioned dynamics reinforce the need for an investor to include a diversifying strategy in order to achieve optimal risk-to-reward ratios in a constantly changing macroeconomic environment. It is vital for an investor to adjust their investment strategy based on the movements in interest rates and other market signals.

Finally, from a macro perspective, the researchers believe future researchers can investigate sophisticated methods such as AI portfolio allocation systems utilizing data analysis for dynamic decision making, and behavioural science, focusing on how investor psychology affects an investor's reaction to changing interest rates.

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